Kuan Chieh (Louis) Lin

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EDUCATION

University of Waterloo — **Waterloo, Canada**
Bachelor of Applied Science, Major in Biomedical Engineering — *Class of 2029*

- **Academics: GPA 3.7/4.0,** Excellent Academic Standing | Scholarship: President's Scholarship of Distinction
- **Technical Skills:** SolidWorks, C++, MATLAB, Simulink, Arduino, Python, Soldering, 3D Printing, HTML, Microsoft Office
- **Certifications:** WHMIS, BioSafety, MATLAB, Simulink, TCPS 2, Machine Shop Cert

RELEVANT EXPERIENCES

Hygea Medical Technology Co., Ltd. — **Beijing, China**
Mechanical Engineering Intern — *Sept 2026 – Present*

- Developing a soft cryoablation needle for natural-orifice treatment of in situ and superficial carcinomas of the bronchial mucosa and submucosa, researching current clinical procedures to fit the device into existing workflows, and mechanically testing R&D prototypes.
- Engineered Python simulations to model theoretical ice ball growth geometries within tissue phantoms, calculating surrounding power dissipation and evaluating phase-change thermodynamics.
- Conducted comparative performance testing on current-generation cryogenic needles in gelatin phantoms to evaluate freezing efficiency, internal temperature gradients, and maximum volumetric size.
- Analyzed experimental thermodynamic data to formulate heat loss equations and determine the effective treatment volume of cryogenic devices, establishing baseline metrics to guide future hardware optimizations.
- Supporting international sales of Hygea's AI CAM capsule endoscopy system to clients in Malaysia, Saudi Arabia and the UAE, presenting the product, answering technical questions and interpreting in client meetings.

Roche Canada — **Mississauga, Canada**
Pharma Technical (Quality) Intern — *Jan 2026 – May 2026*

- Engineered a custom mobile application for the PTQ Global Leaders' Summit 2026, integrating a social networking interface for executive peer-to-peer engagement; developed backend logic for real-time data synchronization, live-voting modules, and dynamic itinerary management.
- Conducted root-cause analysis on SUSVIMO implant complaints, serving as a technical supplier liaison for Phillips-Medisize to evaluate hardware compliance; performed critical failure analysis to determine the necessity of deeper investigations or regulatory recalls.
- Developed an automated data pipeline and logic-based workflow between Google Sheets and Monday.com API to eliminate manual intake latency; authored standardized operating procedures (SOPs) and video modules to facilitate technical onboarding and process scalability.

Premier Healthcare Solutions — **Taipei, Taiwan**
Medical Machinery Summer Intern — *May 2024 – August 2024*

- Monitored Varian's V-PROBE cryoprobe and provided technical support during surgeries
- Researched Varian products, including microsphere technologies, to gain a deeper understanding of their role in advancing targeted cancer therapies and modern oncology practices.
- Maintained and troubleshot Varian's V-PROBE machines through inspections, calibrations, and diagnostics, ensuring reliable performance during critical medical procedures.

UW Biomechatronics Club — **Waterloo, Canada**
Exo-Project Mechanical Team Lead — *September 2024 – Present*

- Leading the Mechanical Subteam in the design and early-stage development of a lower-body exoskeleton for the ACE 2026 Competition, drafting CAD models for a custom cycloidal gearbox and planning mechanical systems with safety and modularity in mind.
- Applied skills from previous robotic hand projects, including 3D printing, assembly, and integration of mechanical components, to develop reliable prototypes and ensure seamless hardware functionality.
- Coordinating with other subteam leads and leveraging Arduino and biosensor experience to guide system integration, iterative design, and multidisciplinary collaboration.

PROJECTS

EMG Controlled Prosthetic Hand — *May 2026 – August 2026*

- Engineered a tendon-driven, myoelectric prosthetic hand prototype using forearm sensors to translate muscle signals into mechanical actuation.
- Programmed an Arduino Uno microcontroller in C++ to process sensor inputs and drive a continuous rotation servo for dynamic grip control.
- Drafted circuit schematics and technical documentation to validate firmware and hardware integration.